



MILITARY CAMOUFLAGED TARGET DETECTION THROUGH MULTIMODAL IMAGES: A COMPARATIVE STUDY

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Abstract

Accurate detection of military camouflaged targets remains a critical challenge in reconnaissance operations. Traditional methods that rely on a single image modality often fail under complex environmental conditions due to camouflage techniques that obscure targets within that spectral domain. This study investigates the detection performance of camouflaged targets using three imaging modalities: long-wave infrared (LWIR), near-infrared (NIR), and visible spectrum (VIS). By analyzing individual sensor performance, we assess detection accuracy across different training/testing scenarios within and across modalities. This enables an evaluation of each modality to determine its relative strengths in capturing spectral and structural cues for target detection. Experiments were conducted across various target types and target signatures (green, gray, and yellow), using the identical training and testing parameters for each modality. To provide a comparative analysis of sensor effectiveness, the detection performance is measured using mean average precision (mAP). Results show that VIS achieves the highest detection rates in most scenarios due to its detailed color and texture information. NIR demonstrates moderate performance and reveals material differences not visible in other bands, though it is sensitive to background complexity. LWIR yields the lowest performance, limited by its lack of texture and color cues despite its thermal contrast advantage. The lowest detection score is consistently obtained for the yellow target signature, indicating stronger camouflage effectiveness, while the green target signature is the most detectable. These findings offer practical insights for sensor selection in reconnaissance systems and highlight the limitations of single modal imaging in complex environments.

Keywords: Camouflage detection, Military camouflaged targets, Multimodal images, MUDCAD-X, YOLO, Deep learning.

Introduction

Military reconnaissance and surveillance missions often encounter challenges in detecting camouflaged targets, as adversaries design concealment strategies to blend objects into their surroundings and disrupt the target's visual, thermal, and structural features, thereby reducing detectability across various sensor types (P. Zhou et al., 2011; Bondžulić et al., 2024; S. Wang et al., 2024). Traditional visual spectrum sensors (VIS) rely on color and texture information, but their effectiveness is limited in complex environments (Y. Liu et al., 2023; Li et al., 2024). Near-infrared (NIR) imaging offers additional discrimination by capturing reflectance properties invisible to the naked eye, which is particularly useful for identifying synthetic materials or vegetation inconsistencies (Hupel & Stütz, 2022; 2023). Thermal infrared (LWIR) sensors provide complementary information by capturing temperature variations; however, they alone may not offer sufficient target discernibility (Hupel & Stütz, 2022; Hupel & Stütz, 2023; Suo et al., 2023).

Prior studies in camouflaged target detection often focus on a single image modality without any comparison with other modalities (Cheng et al., 2021; Putatunda et al., 2022; Van der Burg et al., 2022; M. Liu & Di, 2023; Y. Wang et al., 2023; Karthiga & Asuntha, 2025). However, analyzing the detection capability of multimodal images can provide capability-specific detection insights in order to enhance visibility and avoid low detection performance, high false alarm rates, and missed detections (Hupel & Stütz, 2023; Brouant et al., 2023; Zhang & Zhu, 2024; Peng et al., 2024). This paper adopts a different perspective by analyzing the detection performance of each modality, LWIR, NIR, and VIS, separately. The goal is to identify which modality provides the most reliable detection under varying camouflage conditions and to characterize which target is the most effectively camouflaged and which color target signature is the most susceptible to detection.

Several studies have explored sensor-based camouflage detection techniques. The authors in (Hupel & Stütz, 2022) developed a hyperspectral anomaly detection method for multimodal imagery called Local Point Density (LPD) and compared it with well-known anomaly detection methods such as RX, LRX, CRD, and AED. In (Hupel & Stütz, 2023), a method was proposed to predict sensor performance for camouflaged target exposure. Hao et al (2004) evaluated data similarity to background and military target features using three background types: woodland, desert, and grassland. Their study assessed camouflage effectiveness using background anomaly detection and deep transfer feature learning.

This study investigates the detection performance of camouflaged targets using three imaging modalities independently: long-wave infrared (LWIR), near-infrared (NIR), and visible spectrum (VIS). By analyzing the performance of individual sensors, LWIR, NIR, and VIS, we assess the operational advantages of multimodal images in detecting concealed objects across different training and testing scenarios within and across modalities. Moreover, we will evaluate detection performance using the publicly available benchmark dataset MUDCAD-X designed for military camouflaged target detection scenarios. Furthermore, the detection performance was measured using standardized object detection metrics and performance variations were analyzed across different target signature (green, gray, yellow). These signatures represent common camouflage patterns used in military scenarios.

This study presents an evaluation of a single-modality sensor in capturing spectral and structural cues for target detection, which will contribute to advancing reconnaissance technologies and enhancing the reliability of automated target detection systems in military operations.

Performance Metrics

Detection performance is evaluated using standard average precision metrics (Ma et al., 2022; F. Zhou et al., 2023): $mAP@0.5$, $mAP@0.7$, $mAP@0.9$, and $mAP@[0.5:0.95]$, where:

$mAP@0.5$, $mAP@0.7$, and $mAP@0.9$ represent the mean average precision at intersection over union (IoU) thresholds of 0.5, 0.7, and 0.9, respectively.

$mAP@[0.5:0.95]$ is the mean of mAP values computed at IoU thresholds from 0.5 to 0.95, in steps of 0.05.

Average precisions $mAP@0.5$, $mAP@0.7$, $mAP@0.9$, and $mAP@[0.5:0.95]$ are defined as follows:

$$mAP@0.5 = \frac{1}{N} \sum_{i=1}^N AP_i, IoU = 0.5 \quad (1)$$

$$mAP@0.5 = \frac{1}{N} \sum_{i=1}^N AP_i, IoU = 0.5 \quad (2)$$

$$mAP@0.7 = \frac{1}{N} \sum_{i=1}^N AP_i, IoU = 0.7 \quad (3)$$

$$mAP@0.9 = \frac{1}{N} \sum_{i=1}^N AP_i, IoU = 0.9 \quad (4)$$

$$mAP@[0.5:0.95] = \frac{1}{N} \sum_{i=1}^N AP_i, IoU \in \{0.5, 0.55, 0.6, \dots, 0.95\} \quad (5)$$

where N represents the number of categories in the multiclass detection task, and AP_i is the average precision of the i -th object category, and can be computed as:

$$AP = \int_0^1 P(R) dR \quad (6)$$

where $P(R)$ is the precision (P) as a function of recall (R).

Dataset Description

This work uses the MUDCAD-X dataset to evaluate the performance of LWIR, NIR, VIS sensors in detecting camouflaged targets (Hupel & Stütz, 2023). MUDCAD-X contains 853 annotated and pixel-aligned multisensor images at a resolution of 512×512 pixels. It is designed for military camouflage detection (Hupel & Stütz, 2023).

The dataset includes 11 camouflaged target types as given in Table 1: net2dgreen, turf, persongreen, personyellow, net3dyellow, tarpgreen, net3dgray, hedge, net3dgreen, tarpgray.

Table 1 Camouflage targets in MUDCAD-X dataset, target signature and instances.

Target signature	Camouflaged Target	Instances	
Green	net2dgreen	331	2191
	turf	309	
	persongreen	614	
	tarpgreen	307	
	hedge	311	
	net3dgreen	319	
Gray	net3dgray	204	379
	tarpgray	102	
	fleece	73	
Yellow	personyellow	568	767
	net3dyellow	199	
Total		3337	

Figure 1 shows examples of camouflaged targets from the MUDCAD-X dataset. The images were captured at two different locations within the test site at the University of the Bundeswehr, Munich. The scenes include diverse environments: grassland, gravel, graveled soil, bushes, trees, concrete, and asphalt roads. Data collection took place in spring, summer, and autumn to ensure seasonal diversity and support reliable evaluation (Hupel & Stütz, 2023).

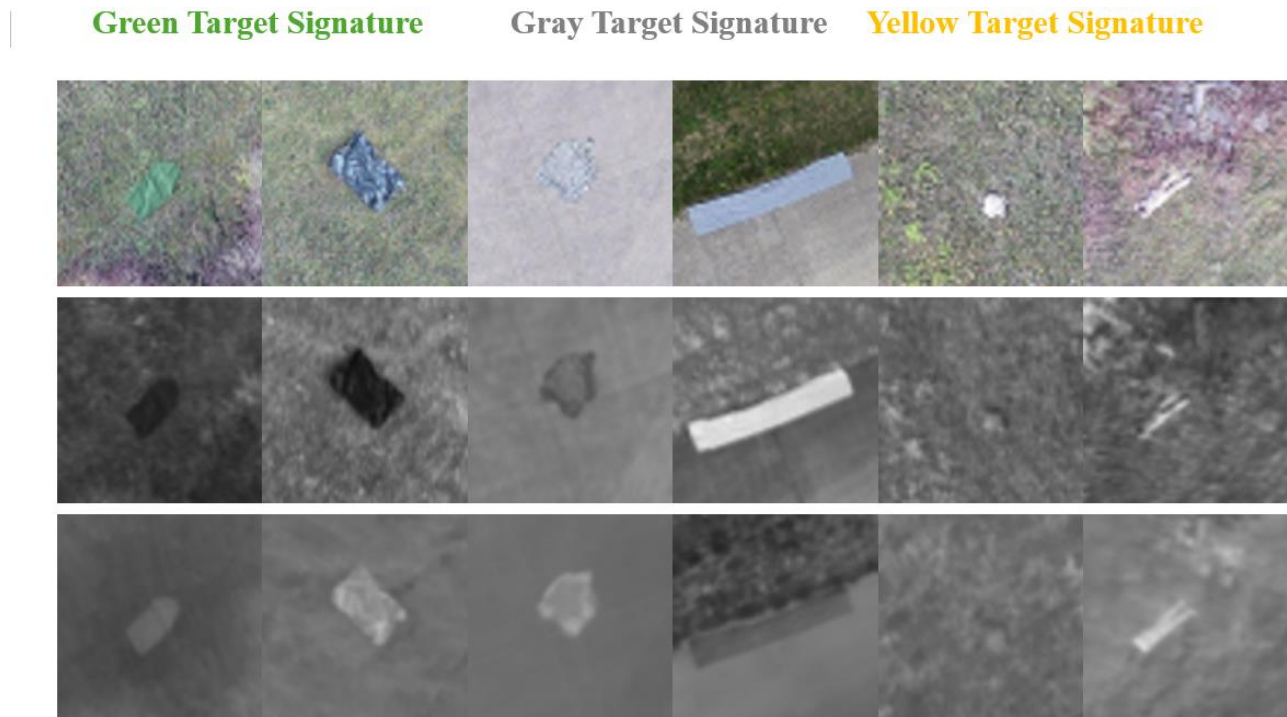


Figure 1 Different classes of the camouflaged targets of the MUDCAD-X dataset.

Experimental Setup

The experiments were performed using the PyTorch framework on a system with an NVIDIA GeForce RTX 3060 GPU and an Intel i7-11800H CPU. The MUDCAD-X dataset was randomly

divided into 80% for training (682 multisensor images) and 20% for testing (171 multisensor images), as illustrated in Figure 2. YOLOv5m with default settings served as the baseline model for camouflage detection. To ensure a fair comparison, separate models were trained on each input type: VIS, LWIR, and NIR. Each model was then tested on the corresponding individual modalities (VIS, LWIR, and NIR).

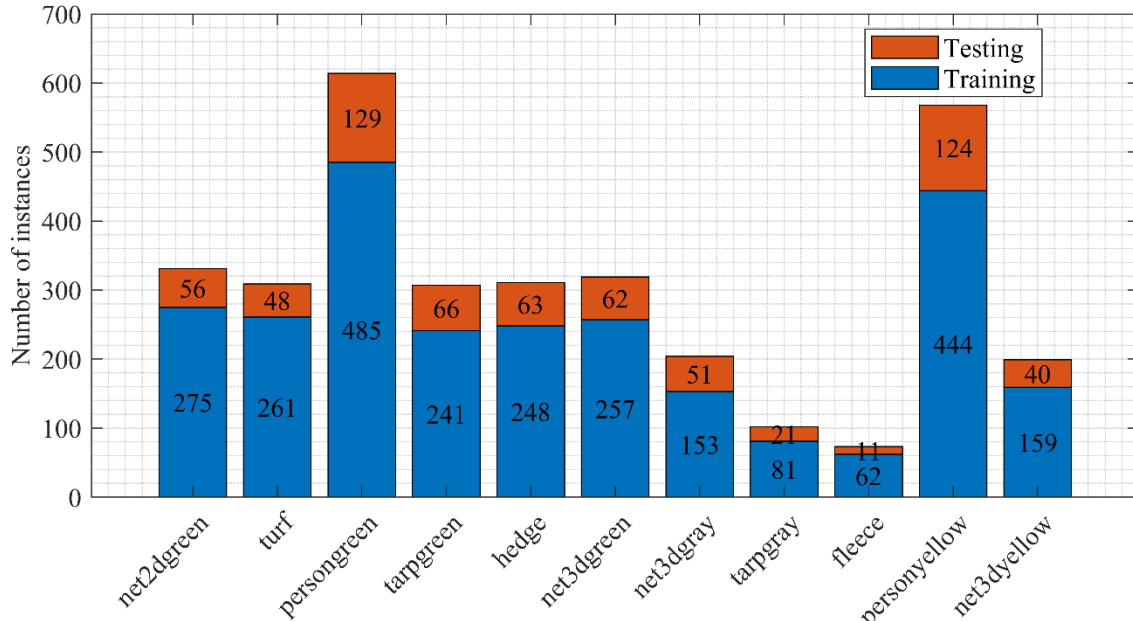


Figure 2 Training/testing partition for MUDCAD-X dataset.

Camouflage Detection Performance across Multimodal Images

To analyze the impact of LWIR, NIR, and VIS bands on camouflage detection, Figure 3 presents the detection performance in terms of mean average precision (mAP) across different training and testing modalities. Each testing set (LWIR, NIR, and VIS) is evaluated using YOLOv5m models trained on various sets, including LWIR, NIR, and VIS bands.

The results show a consistent decline in mAP as the IoU threshold increases from 0.5 to 0.9, highlighting the increasing difficulty of achieving precise detections at higher thresholds. For the LWIR, NIR, and VIS testing sets, the best-performing model is the one trained on the same image modality.

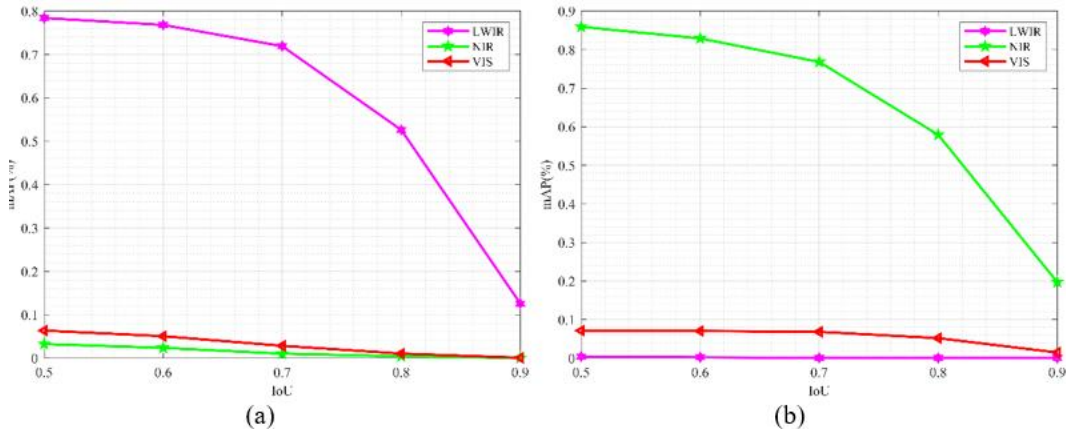


Figure 3 Camouflage detection performance with different training and testing modalities: (a) LWIR, (b) NIR, and (c) VIS testing set.

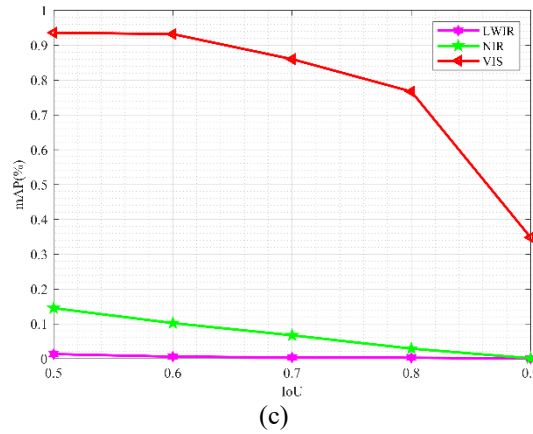


Figure 3 (Continued) Camouflage detection performance with different training and testing modalities: (a) LWIR, (b) NIR, and (c) VIS testing set.

Evaluation by all Targets Combined with Same-Modality Training and Testing

The quantitative results in Table 2 highlight the camouflage detection performance across the LWIR, NIR, and VIS bands. The evaluation uses recall (R), precision (P), and mean average precision (mAP) across different thresholds.

VIS images outperform both individual modalities, achieving the highest $mAP@[0.5:0.95]$ of 0.719. This result highlights the importance of color-based feature information in the VIS spectra, underscoring both thermal contrast and texture information of LWIR and NIR images. It can be also noticed that NIR achieves the second-best performance in terms of mAP, indicating its precise detection capability. In contrast, LWIR performs worse in terms of mAP due to its lack of texture and reliance on thermal contrast, which can be less distinctive for certain targets. LWIR images provide the second best performance in terms of precision and recall, indicating its resilience to false alarm rates and missed detections compared to NIR images.

Table 2 Camouflage detection quantitative performance comparison on LWIR, NIR, and VIS images.

Image	R	P	$mAP@0.5$	$mAP@0.7$	$mAP@0.9$	$mAP@[0.5:0.95]$
LWIR	0.823	0.802	0.784	0.719	0.126	0.545
NIR	0.686	0.785	0.859	0.768	0.197	0.601
VIS	0.84	0.984	0.936	0.86	0.348	0.719

Evaluation by Target Signature with Same-Modality Training and Testing

Table 3 presents $mAP@[0.5:0.95]$ camouflage detection performance for the green, gray, yellow target group. The $mAP@[0.5:0.95]$ represents the average mAP values calculated across IoU thresholds ranging from 0.5 to 0.95, with a step size of 0.05. The results indicate that VIS imagery demonstrates strong performance in camouflage detection. Additionally, the most informative modalities rankings for camouflage detection are as follows: VIS, NIR, LWIR. Moreover, the yellow target group exhibits the lowest detection performance in terms of mAP, suggesting that it is the most effective for camouflage, while the green target group exhibits the best detection performance, indicating that it is the most susceptible to detection. Subsequently, the target signature ranking from most susceptible to least susceptible detection is: (1) green target, (2) gray target, and (3) yellow target.

Table 3 Camouflage detection performance $mAP@[0.5:0.95]$ by target signature.

Target group	Green target	Gray target	Yellow target
LWIR	0,641	0,438	0,418
NIR	0,662	0,599	0,424
VIS	0,787	0,647	0,626

Evaluation by Camouflaged Target Type with Same-Modality Training and Testing

In order to show the strengths and limitations of each multimodal image, LWIR, NIR, and VIS, a comparative analysis of mAP by target is conducted. Figure 4 illustrates the camouflage detection performance in terms of $mAP@[0.5:0.95]$ across different targets for each modality. The black dashed line represents the mean performance of all modalities to highlight the detectability characteristics of each target.

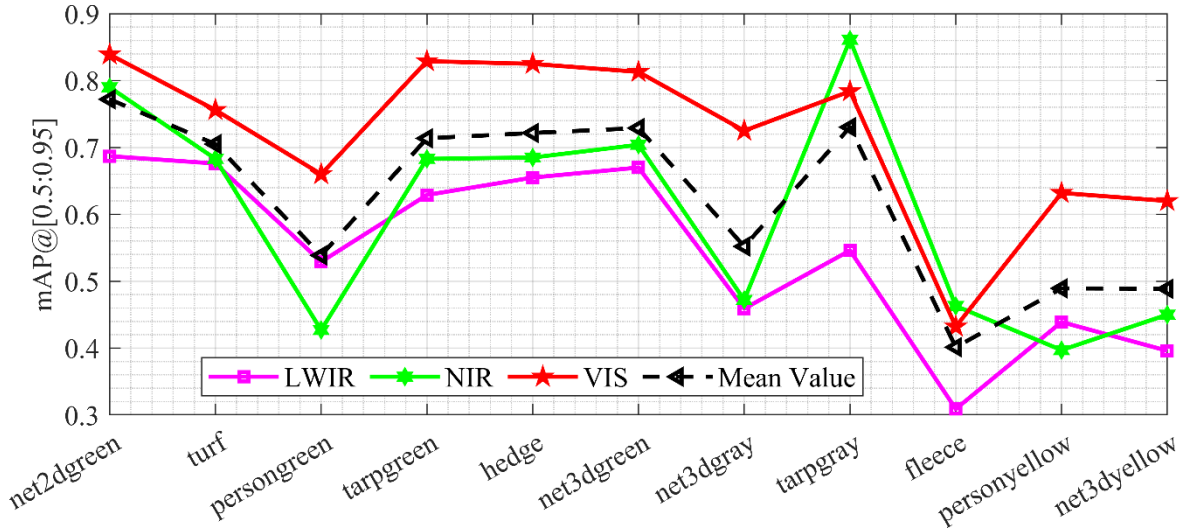


Figure 4 Camouflage detection performance by target type.

It is observed that the LWIR band demonstrates the lowest detection performance compared to NIR and VIS. On the other hand, fusion VIS band achieves the highest performance across almost all targets, reinforcing the advantage of color and texture information, making it the most reliable choice for target detection. However, it is noted that for some targets un the gray group, NIR performs the best, especially, for *tarpgray* and *fleece*.

Based on mean mAP across all single modalities, the three most detectable targets are *net2dgreen* (green target group), *tarpgray* (gray target group) and *net3dgreen* (green target group). The hardest-to-detect targets are *fleece* (gray target group), *personyellow* (yellow target group), and *net3dyellow* (yellow target group).

Subjective Observations of Detection Performance

Figure 5 presents two examples of camouflage detection using LWIR, NIR, and VIS images, illustrating differences in detection confidence. Each example compares individual modalities, highlighting their effectiveness in identifying camouflaged targets.

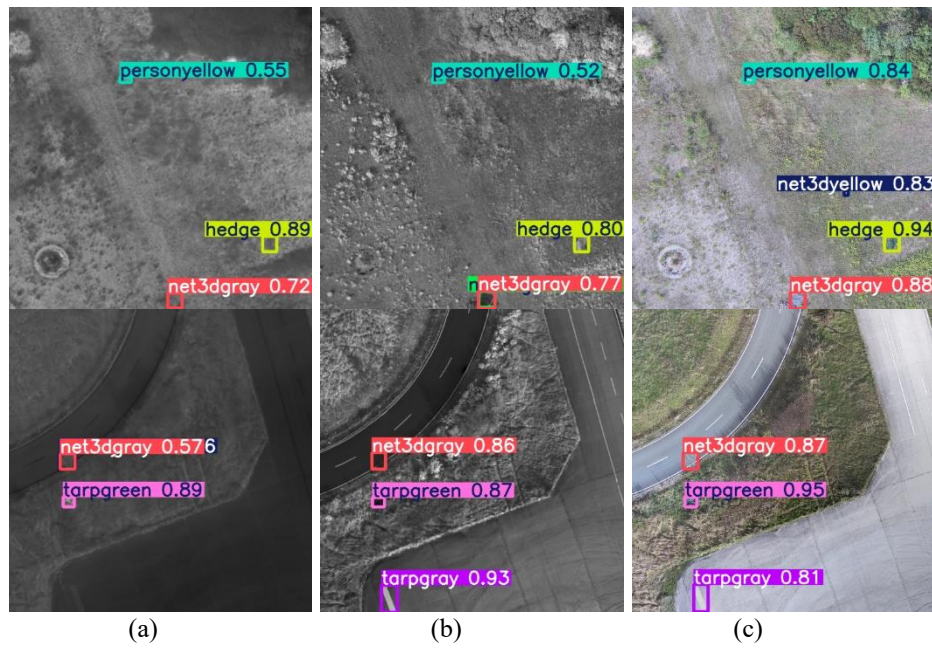


Figure 5 Examples of camouflage detection with same-modality training and testing on identical captures: (a) LWIR, (b) NIR, (c) VIS.

In the first example, VIS achieves the highest detection confidence for all detected camouflaged targets, demonstrating the advantage of color information. Additionally, the LWIR and NIR modalities fail to detect the *net3dyellow* camouflage.

In the second example, VIS outperforms both LWIR and NIR modalities in detecting almost all targets. NIR alone shows the highest detection confidence of *tarpgray* target indicating its discriminative capability in capturing texture and reflectance properties invisible to the naked eye.

Conclusion and Future Work

This study presented a comparative analysis of camouflaged target detection using three imaging modalities: LWIR, NIR, and VIS. Leveraging the MUDCAD-X dataset and a consistent deep learning framework, we evaluated the detection accuracy of each modality under identical training and testing conditions. The results show that the VIS modality provides the highest detection performance due to its ability to capture detailed texture and color information. NIR achieves moderate results by revealing material-specific differences, though it is more affected by complex backgrounds. LWIR performs the worst, limited by its lack of structural detail despite its ability to capture thermal contrast. Detection performance also varies across target signatures. Yellow targets consistently produce the lowest detection scores, indicating a more effective camouflage design across all spectral bands. Green targets are the most easily detected, likely due to stronger contrast with background elements.

Future research should investigate fusion-based approaches that integrate complementary information from multiple spectral domains to enhance detection performance in complex operational environments. Such fusion methods could leverage the texture detail of the VIS and NIR bands to improve detection performance, while using the thermal contrast of the LWIR band to enhance target/background separation and increase resilience to false alarms and missed detections.

References

1. Bondžulić, B., Stojanović, N., Lukin, V., Stankevich, S. A., Bujaković, D., & Kryvenko, S. (2024). Target acquisition performance in the presence of JPEG image compression. *Defence Technology*, 33, 30–41. <https://doi.org/10.1016/j.dt.2023.12.006>
2. Brouant, L., Pham, H.-L., Foucher, É., & Picard, A. (2023). VIS-NIR multispectral for camouflage detection. In R. C. Hollins, G. S. Buller, R. A. Lamb, M. Laurenzis, P. M. Alsing, P. Walther, M. L. Fanto, N. A. Salmon, & F. Gumbmann (Eds.), *Emerging Imaging and Sensing Technologies for Security and Defence VIII* (p. 3). SPIE. <https://doi.org/10.1117/12.2678661>
3. Cheng, X., Shu, B., Chang, Y., Li, X., & Yu, D. (2021). Evaluation of infrared camouflage effectiveness via a multi-fractal method. *Defence Technology*, 17(3), 748–754. <https://doi.org/10.1016/j.dt.2020.05.006>
4. Hao, B., Xu, W., Yang, X., Xiao, F., Li, H., & Huang, W. (2024). Research on military targets camouflage effect rating based on background anomaly detection and deep transfer features learning. *Journal of Industrial Textiles*, 54. <https://doi.org/10.1177/15280837241287937>
5. Hupel, T., & Stütz, P. (2022). Adopting hyperspectral anomaly detection for near real-time camouflage detection in multispectral imagery. *Remote Sensing*, 14(15), 3755. <https://doi.org/10.3390/rs14153755>
6. Hupel, T., & Stütz, P. (2023). Measuring and predicting sensor performance for camouflage detection in multispectral imagery. *Sensors*, 23(19), 8025. <https://doi.org/10.3390/s23198025>
7. Karthiga, K., & Asuntha, A. (2025). CAMOUFLAGE-Net: Comprehensive advanced model for optimal camouflaged target detection and analysis using groundbreaking elements. *Signal, Image and Video Processing*, 19(1), 49. <https://doi.org/10.1007/s11760-024-03580-9>
8. Li, B., Zhou, R., Yang, L., Wang, Q., & Chen, H. (2024). MilDettr: Detection transformer for military camouflaged target detection. *IEEE Access*, 12, 26163–26174. <https://doi.org/10.1109/ACCESS.2024.3363442>
9. Liu, M., & Di, X. (2023). Extraordinary MHNet: Military high-level camouflage object detection network and dataset. *Neurocomputing*, 549, 126466. <https://doi.org/10.1016/j.neucom.2023.126466>
10. Liu, Y., Wang, C., & Zhou, Y. (2023). Camouflaged people detection based on a semi-supervised search identification network. *Defence Technology*, 21, 176–183. <https://doi.org/10.1016/j.dt.2021.09.004>
11. Ma, J., Tang, L., Fan, F., Huang, J., Mei, X., & Ma, Y. (2022). SwinFusion: Cross-domain long-range learning for general image fusion via swin transformer. *IEEE/CAA Journal of Automatica Sinica*, 9(7), 1200–1217. <https://doi.org/10.1109/JAS.2022.105686>
12. Peng, R., Lai, J., Yang, X., Sun, D., Tan, S., Song, Y., & Guo, W. (2024). Camouflaged target detection based on multimodal image input pixel-level fusion. *Frontiers of Information Technology & Electronic Engineering*, 25(9), 1226–1239. <https://doi.org/10.1631/FITEE.2300503>
13. Putatunda, R., Gangopadhyay, A., Erbacher, R. F., & Busart, C. (2022). Camouflaged object detection system at the edge. In K. Jaskie, T. L. Overman, R. I. Hammoud, & A. Mahalanobis (Eds.), *Proc. SPIE 12096, Automatic Target Recognition XXXII*, 120960I (31 May 2022). <https://doi.org/10.1117/12.2618869>
14. Suo, J., Wang, T., Zhang, X., Chen, H., Zhou, W., & Shi, W. (2023). HIT-UAV: A high-altitude infrared thermal dataset for unmanned aerial vehicle-based object detection. *Scientific Data*, 10(1), 227. <https://doi.org/10.1038/s41597-023-02066-6>
15. Van der Burg, E., Hogervorst, M. A., & Toet, A. (2022). Measuring the dynamics of camouflage in natural scenes using convolutional neural networks. In *Proc. SPIE 12270, Target and Background Signatures VIII*, 1227003 (2 November 2022). <https://doi.org/10.1117/12.2636107>
16. Wang, S., Zeng, D., Xu, Y., Yang, G., Huang, F., & Chen, L. (2024). Towards complex scenes: A deep learning-based camouflaged people detection method for snapshot multispectral images. *Defence Technology*, 34, 269–281. <https://doi.org/10.1016/j.dt.2023.12.011>
17. Wang, Y., Fang, Z., Zheng, Y., Yang, Z., Tong, W., & Cao, T. (2023). Dual attribute adversarial camouflage toward camouflaged object detection. *Defence Technology*, 22, 166–175. <https://doi.org/10.1016/j.dt.2021.12.003>
18. Zhang, N., & Zhu, W. (2024). Enhancing target detection accuracy through cross-modal spatial perception and dual-modality fusion. *Frontiers in Physics*, 12. <https://doi.org/10.3389/fphy.2024.1398678>

19. Zhou, F., Deng, H., Xu, Q., & Lan, X. (2023). CNTR-YOLO: Improved YOLOv5 based on ConvNeXt and transformer for aircraft detection in remote sensing images. *Electronics*, 12(12), 2671. <https://doi.org/10.3390/electronics12122671>
20. Zhou, P., Wang, F., Zhang, H., & Xue, M. (2011). Camouflaged target detection based on visible and near infrared polarimetric imagery fusion. In *Proc. SPIE 8194, International Symposium on Photoelectronic Detection and Imaging 2011: Advances in Imaging Detectors and Applications*. <https://doi.org/10.1117/12.899590>